Properties of Continuous-time fourier series:

① Linearity:

$$x(t) \xleftrightarrow{FS} a_k$$

$$y(t) \xleftrightarrow{FS} b_k$$

$$z(t) = Ax(t) + By(t) \xleftrightarrow{FS} Aa_k + Bb_k$$

② Time-shifting:

$$x(t) \xleftrightarrow{FS} a_k$$

$$b_k = \frac{1}{T}\int_T x(t-t_0)e^{-j\omega_0 k t}\,dt$$

Magnitude of the coefficients remain unaltered

Let $t - t_0 = p \Rightarrow dt = dp$

$$\therefore \quad b_k = \frac{1}{T}\int_T x(p)e^{-j\omega_0 k(p+t_0)}\,dt = e^{-j\omega_0 t_0 k}\,a_k$$

$$x(t-t_0) \xleftrightarrow{FS} e^{-j\omega_0 t_0 k}\,a_k$$

③ Time-reversal:

$$x(-t) \xleftrightarrow{FS} a_k$$

$$b_k = \frac{1}{T}\int_T x(-t)e^{-j\omega_0 k t}\,dt = a_{-k}$$

If $x(-t) = x(t)$ then $a_{-k} = a_k$.

If $x(-t) = -x(t)$ then $a_{-k} = -a_k$.

④ Time-scaling

Time scaling changes the period of a periodic signal. If $x(t)$ is a periodic signal with period T, then $x(\alpha t)$ is periodic with period T/α. and fundamental frequency $\alpha\omega_0$. Hence the co-efficients donot change but the co-efficients corresponds to different frequency now.

$$x(t) \xleftrightarrow{FS} a_k$$

$$\therefore\ x(\alpha t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk(\alpha\omega_0)t}$$

⑤ Multiplication

Suppose that $x(t)$ and $y(t)$ are bothe periodic with period T and that

$$x(t) \xleftrightarrow{FS} a_k$$

$$y(t) \xleftrightarrow{FS} b_k$$

Then

$$x(t)\,y(t) \xleftrightarrow{FS} \sum_{l=-\infty}^{\infty} a_l\, b_{k-l} = h_k$$

⑥ Conjugate Symmetry and Conjugation:

$$x(t) \xleftrightarrow{FS} a_k$$

$$x^*(t) \xleftrightarrow{FS} a^*_{-k}$$

For real-valued signal

$x(t) = x^*(t)$

$a_{-k} = a_k^*$.

$\Downarrow$

a_0 is real

$|a_k| = |a_{-k}|$

$x(t)$ is real & even

$a_k = a_{-k} = a_k^*$

⑦ Differentiation

$$\frac{d}{dt} x(t) \xleftrightarrow{FS} j\omega_0 k \, a_k$$

⑧ Integration

$$\int_{-\infty}^{t} x(t)\,dt < \infty \xleftrightarrow{FS} \frac{1}{j\omega_0 k} a_k.$$

Parseval's Relation:

$$\underbrace{\frac{1}{T}\int_T |x(t)|^2 dt}_{\text{Average power in one period}} = \sum_{k=-\infty}^{\infty} |a_k|^2$$

Further, $$\underbrace{\frac{1}{T}\int_T |a_k e^{jk\omega_0 t}|^2 dt}_{\text{Average power in the k-th harmonic component of } x(t).} = \frac{1}{T}\int_T |a_k|^2 dt = |a_k|^2$$

Total average power in a periodic signal	=	Sum of the average power in all of it's harmonic components.

Proof: The fourier series co-efficients of $x(t)$ are a_k and that of $x^*(t)$ are $a^*_{-k} =: b_k$

$\therefore$ The fourier series co-efficients of $x(t)x^*(t)$ are

$$\sum_{l=-\infty}^{\infty} a_l b_{k-l} = \sum_{l=-\infty}^{\infty} a_l a^*_{l-k} =: c_k$$

$$\therefore \quad \frac{1}{T}\int_T x(t)\,x^*(t)\,e^{-j\omega_0 k t}\,dt = c_k$$

$$\Rightarrow \quad \frac{1}{T}\int_T |x(t)|^2 e^{-j\omega_0 k t}\,dt = c_k$$

At $k=0$, $\quad \frac{1}{T}\int_T |x(t)|^2\,dt = c_0 = \sum_{l=-\infty}^{\infty} a_l a_l^*$

$$= \sum_{k=-\infty}^{\infty} |a_k|^2.$$